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September 24, 2026

To:

The Editor-in-Chief
Computer Methods and Programs in Biomedicine
Elsevier

Subject: Submission of Original Research Article

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript entitled:

“Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Data Cohort Shift”

for publication consideration as an **Original Research Article** in *Computer Methods and Programs in Biomedicine*.

Significance and Alignment with CMPB Scope:

Clinical deep learning algorithms for chest radiography increasingly suffer from severe probability miscalibration when deployed over time. In biomedical image classification, positive-class weighting within binary cross-entropy (BCE) is standard practice to manage acute class imbalance (prevalence $< 5\%$). However, its algebraic interaction with model output logits has been widely conflated with intrinsic clinical feature drift.

Our study directly addresses the algorithmic and computational foundations of this failure:

- 1. Mathematical Derivation & Empirical Clustering:** We derive that positive loss weighting mathematically induces an additive constant logit displacement of $+\ln w$ in the population Bayes-optimal classifier. Across 16,604 evaluations spanning DenseNet-121 and ResNet-50 backbones on 40,523 radiographs from the BRAX benchmark, finite neural network logits cluster tightly around theoretical $\ln w$ (3.29 ± 0.81 vs. theoretical 3.07).
- 2. Structural Failure of Temperature Scaling:** We prove that standard temperature scaling (z/T) modifies logit variance without shifting the intercept, rendering it structurally incapable of correcting this intercept displacement ($\alpha \approx -2.96$).
- 3. Algorithmic $O(1)$ Analytic Recalibration:** We formulate a closed-form analytic logit offset ($z - \ln w$) that requires zero numerical optimization, zero validation tuning, and $O(1)$ constant

compute, eliminating 73.7% of stratum T_3 Brier error ($0.1079 \rightarrow 0.0284$, $\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]) and restoring calibration parity without model retraining.

4. **Budget-Ranked Selective Deferral:** We formalize the actual uncertainty-ranked selective triage pipeline, quantifying the clinical trade-off between calibrated risk estimation and recall-oriented screening referral.
5. **External Multi-Center Transportability Stress-Test:** We evaluate external validity on the official MIMIC-CXR-JPG frontal test split (3,403 images from 3,041 studies across 289 patients), establishing both the efficacy and the boundaries of zero-retraining recalibration under real-world domain shift.

Author Declarations and Compliance:

- This manuscript represents original work and has not been published previously, nor is it under simultaneous consideration for publication elsewhere.
- Publication is approved by the author and the host institution.
- No funding was received for this study, and the author declares no competing financial or non-financial interests.
- All analyzed benchmark cohorts (BRAX v1.1.0 and MIMIC-CXR-JPG v2.1.0) are de-identified, credentialed-access research datasets compliant with institutional ethics review and HIPAA standards.
- Complete analysis code and evaluation scripts supporting all findings are documented for deterministic replication.

Thank you for your time and consideration of our manuscript.

Sincerely,

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